SELF-REFLECTIVE REPORT

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Project: Pastimes Second-Hand Branded Clothing Web Application

INTRODUCTION

The Portfolio of Evidence (PoE) project for WEDE6021 was a comprehensive web development project that required the application of both technical and professional skills. The objective of the project was to design and develop a fully functional web-based platform called Pastimes, a second-hand branded clothing marketplace that allows users to buy and sell clothing items through a secure and user-friendly online environment. The project involved planning, research, database development, coding, testing, documentation, and teamwork. Throughout the project, I gained valuable experience in software development while learning how different components of a web application work together to provide a complete solution for users. The PoE challenged me to think critically, solve problems independently, work collaboratively with team members, and apply theoretical concepts learned during the module to a real-world scenario. The project also provided insight into professional software development practices, including requirement analysis, system implementation, debugging, and project documentation. Overall, the PoE was a valuable learning experience that helped me improve my technical skills and better understand the software development life cycle.

ROLE IN THE TEAM

During the development of the Pastimes project, I contributed to multiple phases of the project lifecycle, taking on different responsibilities across all three parts of the Portfolio of Evidence.

Part 1: Research, Planning and Design

In Part 1, I was responsible for **Section 2: Planning and Design** of the web application. This involved:

- Conducting research on existing second-hand clothing platforms to understand their features and functionality
- Researching how courier companies integrate with e-commerce platforms for delivery coordination
- Investigating how websites integrate with banks for payment processing and order management
- Studying how parcel tracking systems work so customers can see where their items are in real time
- Designing the complete system architecture and creating detailed UI mock-ups
- Developing the database schema and ERD diagrams
- Creating the project plan with deadlines and milestones
- Documenting all functions and user interactions

This research was critical because it helped us understand how real-world e-commerce platforms operate and what features we needed to include in Pastimes. Learning about courier integration, payment systems, and parcel tracking informed our design decisions and ensured we built a practical, functional application.

Part 2: Prototype Implementation

In Part 2, I focused on the following areas:

Database Development:

- Wrote the SQL code to create the complete database structure
- Created all tables with proper relationships, primary keys, and foreign keys
- Populated the database with sample data using SQL scripts
- Ensured data integrity and proper constraints

Testing and Debugging:

- Tested the code thoroughly to identify and fix minor bugs
- Improved existing code by adding features where I thought they were needed

- Ensured the application functioned correctly across different scenarios

Documentation:

- Added comments to the code to explain functionality and improve readability
- Documented the Part 2 implementation process
- Created comprehensive documentation for the project

Video Demonstration:

- Created the demonstration video showing the application's functionality
- Walked through all key features including registration, login, admin verification, and shopping

Part 3: New Features Implementation

In Part 3, I continued to contribute to the project:

Database Enhancements:

- Wrote additional SQL code to support new features
- Added new items to the database including product images
- Extended the database structure to accommodate seller requests and tracking functionality

Testing and Quality Assurance:

- Continued testing the code and fixing bugs
- Ensured new features integrated smoothly with existing functionality

Video Demonstration:

- Created the Part 3 demonstration video showcasing the new features

Collaboration and Communication

My collaboration with my teammate, Fortunate Lukhele, was highly effective. We had the advantage of living in the same complex, which made it easy to meet frequently and discuss the project. This proximity allowed us to:

- Have regular face-to-face meetings to discuss progress
- Work together on challenging technical problems
- Share ideas and provide immediate feedback
- Coordinate our efforts effectively

Communication played an important role in ensuring that tasks were completed effectively. We regularly discussed progress, shared ideas, and provided feedback to each other through WhatsApp and in-person meetings. Whenever challenges arose, we worked together to find practical solutions and support one another. We also used Google Drive to share documents and maintain a central repository of project files.

Research Responsibilities

Research was a critical component of the project because it enabled us to make informed decisions during planning and development. I conducted research in several key areas:

Scenario 1: Researching Existing Platforms
I researched existing second-hand clothing platforms like Yaga and Love It Again to understand their features, strengths, and weaknesses. This research helped me understand what makes a successful online marketplace and which features to incorporate into Pastimes.

Scenario 2: Researching Courier Integration
I researched how courier companies integrate with e-commerce platforms for delivery coordination. This included understanding how tracking numbers are generated, how delivery status is updated, and how customers can track their parcels in real time.

Scenario 3: Researching Payment Integration
I investigated how websites integrate with banks for payment processing and order management. This helped us understand the requirements for processing transactions securely.

Scenario 4: Researching Technical Solutions
When implementing the user verification workflow, I needed to research how to add status columns to the database and check them during login. I also researched sticky form implementation, session management, and secure password hashing.

To resolve these challenges, I consulted online tutorials, PHP documentation, educational websites, and developer forums. Through research, I learned how to implement secure

login systems, handle user sessions effectively, validate user input, and improve overall application functionality.

PERSONAL STRENGTHS AND WEAKNESSES

Tasks I Performed Well

1. Database Development and SQL Coding
I successfully wrote SQL code to create the complete database structure with proper relationships, primary keys, and foreign keys. I also added new items to the database including product images, ensuring the database was comprehensive and well-organized. My ability to work with databases allowed me to add rows, columns, and items efficiently.

2. Testing and Debugging
I was good at testing the code and fixing minor bugs that were present. I had a keen eye for identifying issues and finding solutions to make the code work properly. Once a code was written, I was good at finding ways to improve it and add more features where I thought they were needed.

3. Code Understanding and Optimization
I am good at understanding code and making it work to its fullest potential. I have the ability to analyze existing code, understand its functionality, and identify opportunities for improvement. This skill allowed me to add features and enhance the application's functionality.

4. Documentation and Commenting
I added comments to the code to explain functionality and improve readability. I also documented the Part 2 implementation process and contributed to the overall project documentation.

5. Video Demonstrations
I created both the Part 2 and Part 3 demonstration videos, walking through the application's functionality and showcasing all key features.

Strengths I Identified

Strength	Description	Evidence
Database Management	Strong ability to design and manage databases, add items, and maintain data integrity	Successfully created and maintained the ClothingStore database
Testing and Debugging	Ability to identify and fix bugs in code	Successfully tested and improved the application code
Code Understanding	Capable of understanding existing code and improving it	Added features and enhancements to the existing codebase
Research Skills	Ability to find relevant information for project requirements	Researched courier integration, payment systems, and technical solutions
Collaboration	Worked effectively with team member through regular meetings	Successfully collaborated despite different skill sets
Documentation	Created clear, comprehensive documentation	Added comments and documented project progress

Tasks I Did Not Perform Well

1. Writing PHP Code

I was not good at writing PHP code from scratch. This was a significant challenge because PHP was the primary language for the backend development. I needed to research a lot of things, which would have cost us a lot of time if I had to do it alone.

2. CSS Styling

I was not proficient in CSS styling. Creating visually appealing interfaces was outside my comfort zone, and I relied on my teammate for this aspect of the project.

3. Thinking on My Feet

I was not good at thinking on my feet when it came to writing code. When faced with new

coding challenges, I needed time to research and understand the problem before finding a solution.

4. Independent Code Creation
I struggled to create code independently without having a reference or example to work from.

Reasons for Underperformance

The primary reason I did not perform as strongly in these areas was that I am still learning PHP and CSS. These are skills that I am developing, and I have not yet achieved proficiency. Without a solid foundation in these languages, I needed to research extensively, which would have been time-consuming. Fortunately, because I worked with a teammate who had complementary skills, we were able to divide the work according to our strengths.

Another factor was that I needed more practice and hands-on experience with these technologies. Coding is a skill that improves with practice, and I have not yet had enough opportunity to practice writing PHP and CSS code independently.

Improvement Plan

To improve these weaknesses, I intend to:

1. Study PHP in Depth
I will take online courses and complete coding exercises to build my PHP skills. I will practice writing PHP code regularly to become more comfortable with the language.

2. Learn CSS and Frontend Design
I will study CSS frameworks like Bootstrap and learn frontend design principles. This will help me contribute to both backend and frontend development in future projects.

3. Practice Coding Regularly
I will set aside time each day to practice coding. The more I practice, the more comfortable I will become with writing code independently.

4. Work on Personal Projects
I will build small projects on my own to practice the skills I am learning. This will give me hands-on experience and build my confidence.

5. Study Code Examples
I will study well-written code examples to understand best practices and learn from experienced developers.

The self-reflection process has allowed me to evaluate my performance honestly and identify opportunities for growth. I now have a clearer understanding of both my strengths and the areas that require further development as I continue progressing in my studies and future career.

CONCLUSION

In conclusion, the WEDE6021 Portfolio of Evidence was a highly rewarding and educational experience that enabled me to apply theoretical knowledge in a practical development environment. My contributions across all three parts of the project allowed me to develop a wide range of skills, from research and planning to database development, testing, and documentation.

My role in Part 1 involved designing the system architecture and creating detailed plans for the application. In Part 2, I focused on SQL coding, testing, debugging, documentation, and video creation. In Part 3, I continued with database enhancements, testing, and another video demonstration. Working alongside Fortunate Lukhele provided valuable experience in teamwork, communication, and collaboration. Our ability to meet frequently because we live in the same complex made collaboration particularly effective.

The project improved my research, problem-solving, technical, and professional skills while highlighting areas for future growth. I recognized that while I am strong in database management, testing, and understanding code, I need to improve my PHP and CSS skills. I plan to address these weaknesses through continued learning and practice.

Overall, the PoE provided a realistic understanding of software development processes and prepared me for future academic and professional challenges within the Information Technology industry. I am grateful for this opportunity and excited about the doors it will open for my future career.